经典力学第六次作业

L07.05

根据几何关系得到

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\omega t & -\sin\omega t \\ \sin\omega t & \cos\omega t \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix},$$

两边同时求导数

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} \cos\omega t & -\sin\omega t \\ \sin\omega t & \cos\omega t \end{pmatrix} \begin{pmatrix} \dot{x}' \\ \dot{y}' \end{pmatrix} - \omega \begin{pmatrix} \sin\omega t & \cos\omega t \\ -\cos\omega t & \sin\omega t \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix},$$

于是

$$\dot{x}^2 + \dot{y}^2 = \begin{pmatrix} \dot{x} & \dot{y} \end{pmatrix} \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \left(\dot{x}^2 + \dot{y}^2\right) + \omega^2 \left(x'^2 + y'^2\right) + 2\omega \left(x'\dot{y}' - \dot{x}'y'\right).$$

在 $Oxyz$ 系中, Lagrange 量写作

$$L = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - V(\boldsymbol{r}),$$

直接代入相关表达式就得到 $Ox'y'z'$ 系的 Lagrange 量

$$L' = \frac{1}{2} m \left[\left(\dot{x}'^2 + \dot{y}'^2 + \dot{z}'^2\right) + \omega^2 \left(x'^2 + y'^2\right) + 2\omega \left(x'\dot{y}' - \dot{x}'y'\right)\right] - V(\boldsymbol{r}'),$$

据此求得广义动量

$$\begin{cases} p_{x'} = \dfrac{\partial L'}{\partial x'} = m\left(\dot{x}' - \omega y'\right), \\ p_{y'} = \dfrac{\partial L'}{\partial y'} = m\left(\dot{y}' + \omega x'\right), \\ p_{z'} = \dfrac{\partial L'}{\partial z'} = m\dot{z}, \end{cases}$$

反解广义速度

$$\begin{cases} \dot{x}' = \dfrac{p_{x'}}{m} + \omega y', \\ \dot{y}' = \dfrac{p_{y'}}{m} - \omega x', \\ \dot{z} = \dfrac{p_{z'}}{m}, \end{cases}$$

对 Lagrange 量做 Legendre 变化得到广义能量函数, 并替换所有的广义速度为广义动量

$$H' = \dot{x}' p_{x'} + \dot{y}' p_{y'} + \dot{z}' p_{z'} - L = \frac{p_{x'}^2 + p_{y'}^2 + p_{z'}^2}{2m} + \omega \left(y' p_x{}' - x' p_{y'}\right) + V(\boldsymbol{r}').$$

L07.09

(1)

依照题意,可以归纳出

$$\boldsymbol{J}=\boldsymbol{r}\times\boldsymbol{p},\qquad \boldsymbol{K}=\boldsymbol{p}\times\boldsymbol{J}+mV(\boldsymbol{r})\boldsymbol{r}.$$

由于 $\boldsymbol{J}$ 的方向沿着 z 轴但是表达式中不含 z,p_z 于是

$$[\boldsymbol{J},H]=\partial_{\boldsymbol{r}}\boldsymbol{J}\partial_{\boldsymbol{p}}H-\partial_{\boldsymbol{p}}\boldsymbol{J}\partial_{\boldsymbol{r}}H=\dot{\boldsymbol{r}}\cdot\partial_{\boldsymbol{r}}\boldsymbol{J}-\dot{\boldsymbol{p}}\cdot\partial_{\boldsymbol{p}}\boldsymbol{J}=\dot{z}\partial_z J_z-\dot{p}_z\partial_{p_z}J_z=0.$$

$$[\boldsymbol{K},H]=\partial_{\boldsymbol{r}}\boldsymbol{K}\partial_{\boldsymbol{p}}H-\partial_{\boldsymbol{p}}\boldsymbol{K}\partial_{\boldsymbol{r}}H=\dot{\boldsymbol{r}}\partial_{\boldsymbol{r}}\boldsymbol{K}-\dot{\boldsymbol{p}}\partial_{\boldsymbol{p}}\boldsymbol{K}=\dot{\boldsymbol{r}}\left(V(\boldsymbol{r})+\boldsymbol{r}\cdot\nabla V(\boldsymbol{r})\right)=\dot{\boldsymbol{r}}\left(-\frac{k}{r}+\boldsymbol{r}\cdot\frac{k}{r^2}\hat{r}\right)=0.$$

这就得证了.

(2)

采用矩阵求导的方法,这样做可以完整地表现各个向量之间的关系.

$$\begin{aligned}
[\boldsymbol{K},\boldsymbol{J}]&=\partial_{\boldsymbol{r}}\boldsymbol{K}(\partial_{\boldsymbol{p}}\boldsymbol{J})^{\mathrm{T}}-\partial_{\boldsymbol{p}}\boldsymbol{K}(\partial_{\boldsymbol{r}}\boldsymbol{J})^{\mathrm{T}}\\
&=\begin{pmatrix}-\frac{mk}{r}+\frac{mkx^2}{r^3} & \frac{mkxy}{r^3} & \frac{mkxz}{r^3}\\ \frac{mkxy}{r^3} & -\frac{mk}{r}+\frac{mky^2}{r^3} & \frac{mkyz}{r^3}\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}0&0&-y\\0&0&x\\0&0&0\end{pmatrix}-\begin{pmatrix}0&J&0\\-J&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}0&0&p_y\\0&0&-p_x\\0&0&0\end{pmatrix}\\
&=\begin{pmatrix}0&0&\frac{mky}{r}\\0&0&-\frac{mkx}{r}\\0&0&0\end{pmatrix}-\begin{pmatrix}0&0&-p_xJ\\0&0&-p_yJ\\0&0&0\end{pmatrix}\\
&=\begin{pmatrix}0&0&-K_y\\0&0&K_x\\0&0&0\end{pmatrix}.
\end{aligned}$$

接下来是最后一个关系的证明,

$$\begin{aligned}
[K_x,K_y]&=\partial_{\boldsymbol{r}}K_x(\partial_{\boldsymbol{p}}K_y)^{\mathrm{T}}-\partial_{\boldsymbol{p}}K_x(\partial_{\boldsymbol{r}}K_y)^{\mathrm{T}}\\
&=\left(p_y^2-\frac{mk}{r}+\frac{mkx^2}{r^3},-p_xp_y+\frac{mkxy}{r^3},\frac{mkxz}{r}\right)\begin{pmatrix}-xp_y+2yp_x\\-xp_x\\0\end{pmatrix}-(-yp_y,2xp_y-yp_x,0)\begin{pmatrix}-p_xp_y+\frac{mkxy}{r^3}\\p_x^2-\frac{mk}{r}+\frac{mky^2}{r^3}\\\frac{mky^2}{r^3}\end{pmatrix}\\
&=(yp_x-xp_y)\left[\left(p_x^2+p_y^2\right)-\frac{2mk}{r}\right]\\
&=-(2mH)J.
\end{aligned}$$

复刻这个手法,在 Poinsson 括号内放入两个角动量矢量,

$$\begin{aligned}
[\boldsymbol{J},\boldsymbol{J}]&=\partial_{\boldsymbol{r}}\boldsymbol{J}(\partial_{\boldsymbol{p}}\boldsymbol{J})^{\mathrm{T}}-\partial_{\boldsymbol{p}}\boldsymbol{J}(\partial_{\boldsymbol{r}}\boldsymbol{J})^{\mathrm{T}}\\
&=\begin{pmatrix}0&p_z&-p_y\\-p_y&0&p_x\\p_y&-p_x&0\end{pmatrix}\begin{pmatrix}0&z&-y\\-z&0&x\\y&-x&0\end{pmatrix}-\begin{pmatrix}0&-z&y\\z&0&-x\\-y&x&0\end{pmatrix}\begin{pmatrix}0&-p_z&p_y\\p_y&0&-p_x\\-p_y&p_x&0\end{pmatrix}\\
&=\begin{pmatrix}0&J_z&-J_y\\-J_z&0&J_x\\J_y&-J_x&0\end{pmatrix}.
\end{aligned}$$

如果记 $M_i=K_x/\sqrt{2m\,|E|},K_y/\sqrt{2m\,|E|},J,i=1,2,3$ 对应 x,y,z, 那么根据之前的结论

$$[M_1,M_2]=\frac{1}{-2mH}[K_x,K_y]=J=M_3,$$

$$[M_2,M_3]=\frac{1}{\sqrt{2m\,|E|}}[K_y,J]=\frac{K_x}{\sqrt{2m\,|E|}}=M_1,$$

$$[M_3, M_1] = -\frac{1}{\sqrt{2m\,|E|}}[K_x, J] = \frac{K_y}{\sqrt{2m\,|E|}} = M_2.$$

\end{align*} $$

(3)

直接计算,

$$\begin{aligned} M_1^2 + M_2^2 + M_3^2 &= \frac{1}{-2mH}\left(K_x^2 + K_y^2 - 2mHJ^2\right) \\ &= \frac{1}{-2mH}\left[\left(p_y J - \frac{mkx}{r}\right)^2 + \left(-p_x J - \frac{mky}{r}\right)^2 - 2mHJ^2\right] \\ &= \frac{mk^2}{2\,|E|}. \end{aligned}$$

J04.05

(A)

The generalized momentum p_k is derived by formula $p_k = \frac{\partial L}{\partial \dot{q}_k}$, so

$$\begin{aligned} p_1 &= 2a\dot{q}_1 + 2b\dot{q}_2, \\ p_2 &= 2b\dot{q}_1 + 2c\dot{q}_2. \end{aligned}$$

And write them as form of matrix,

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = 2\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix},$$

so

$$\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} = \frac{1}{2}\frac{1}{ac - b^2}\begin{pmatrix} c & -b \\ -b & a \end{pmatrix}\begin{pmatrix} p_1 \\ p_2 \end{pmatrix}.$$

That is

$$\begin{aligned} \dot{q}_1 &= \frac{cp_1 - bp_2}{2\,(ac - b^2)}, \\ \dot{q}_2 &= \frac{-bp_1 + ap_2}{2\,(ac - b^2)}. \end{aligned}$$

(B)

The H is the L's Legendre transformation, so

$$H = p_1\dot{q}_1 + p_2\dot{q}_2 - L = a\dot{q}_1^2 + 2b\dot{q}_1\dot{q}_2 + c\dot{q}_2^2 - f.$$

Subsitute the $\dot{q}$ with p, and we have

$$H = \frac{1}{4\,(ac - b^2)}\left(cp_1^2 - 2bp_1p_2 + ap_2^2\right).$$

(C)

Take it into Hamilton's equation,

$$\dot{q}_1 = \frac{\partial H}{\partial p_1} = \frac{cp_1 - bp_2}{2\,(ac - b^2)}, \qquad \dot{q}_2 = \frac{\partial H}{\partial p_2} = \frac{-bp_1 + ap_2}{2\,(ac - b^2)}.$$

So we verified that!

(D)

Just like in Lagrange's Mechanics that the $\dot{q}$ is independent from t and q, here, in Hamilton's Mechanics we subsitute $\dot{q}$ with p, and naturally the p is independent.

J04.10

(A)

Use only one step $[f,p]=\partial_q f\partial_p p-\partial_p f\partial_q p=\partial_q f$ and QED.
Detailly, the q is a vector here, and with $\partial_p p=1$, $\partial_q p=0$, so

$$[f,p]=\partial_q f=\begin{pmatrix}\partial_x f & \partial_y f & \partial_z f\end{pmatrix}^{\mathrm{T}}.$$

(B)

Just like what we do above, we do again,

$$\begin{aligned}[L_z,q]&=\partial_q L_z\partial_p q-\partial_p L_z\partial_q q=-\partial_p L_z=-\begin{pmatrix}\partial_{p_x} & \partial_{p_y} & \partial_{p_z}\end{pmatrix}^{\mathrm{T}}(xp_y-yp_x)\\&=\begin{pmatrix}y & -x & 0\end{pmatrix}^{\mathrm{T}},\end{aligned}$$

$$\begin{aligned}[L_z,p]&=\partial_q L_z\partial_p p-\partial_p L_z\partial_q p=\partial_q L_z=\begin{pmatrix}\partial_x & \partial_y & \partial_z\end{pmatrix}^{\mathrm{T}}(xp_y-yp_x)\\&=\begin{pmatrix}p_y & -p_x & 0\end{pmatrix}^{\mathrm{T}}.\end{aligned}$$

J04.12

(A)

Subsitute instinctly,

$$\begin{aligned}&[Q_k,Q_l]=[p_k,p_l]=0,\\&[Q_k,P_l]=[p_k,-q_l]=[q_l,p_k]=\delta_{kl},\\&[P_k,P_l]=[-q_k,-q_l]=0.\end{aligned}$$

(B)

Subsitute instinctly,

$$H(P,Q,t)=\frac{X^2+Y^2+Z^2}{2m}+U(-P_X,-P_Y,-P_Z,t).$$

(C)

In q,p system,

$$\dot{q}=\frac{\partial H}{\partial p}=\frac{p}{m},\quad \dot{p}=-\frac{\partial H}{\partial q}=-\nabla_q U,\quad \dot{H}=\frac{\partial H}{\partial t}=\dot{U}.$$

In Q,P system,

$$\dot{Q}=\frac{\partial H}{\partial P}=\nabla_P U,\quad \dot{P}=-\frac{\partial H}{\partial Q}=-\frac{Q}{m},\quad \dot{H}=\frac{\partial H}{\partial t}=\dot{U}.$$

And noticed,

$$\dot{p}=-\nabla_q U,\quad -\dot{q}=-\frac{p}{m},$$

so the equation formation is equivalent in this transformation.